Experiment 10

Aim

To test and review the Campus Connect Flutter application, configure an Android upload keystore, and prepare the release signing and deployment workflow.

Theory

Flutter applications should be analysed and tested before deployment. Static analysis checks Dart code, widget tests check interface behavior, and runtime testing examines the application on its target device. APK files support Android installation, while Android App Bundles support store distribution.

Android release builds require a signing identity. A keystore holds the private key, key.properties supplies its local configuration, and Gradle applies the release signing configuration. The key password and private key must remain private.

Steps

1 Test the application Inspect available devices, run the application, and check form validation, Firebase save and read actions, search, filtering, and deletion confirmation.

2 Debug and review Run flutter analyze and the setup widget test. Review the Android toolchain with flutter doctor -v.

3 Create the upload keystore Generate an RSA upload key using keytool and inspect its certificate. Store the password in the ignored local key.properties file.

4 Configure release signing Load key.properties in android/app/build.gradle.kts and assign the release signing configuration to release builds.

5 Build and prepare deployment Generate and validate the release APK and App Bundle once the Android toolchain is complete. Prepare the store listing and test on Android before publishing.

Output Terminal Testing

Commands executed in the campus_connect project. The images below display captured command output; they were rendered directly from the saved terminal logs.

Campus Connect | PowerShell command output

```
PS campus_connect> flutter analyze
Analyzing campus_connect...
No issues found! (ran in 4.5s)

PS campus_connect> flutter test test/setup_test.dart
00:00 +0: loading D:/Campus_Connect_Flutter_Firebase/campus_connect/test/setup_test.dart
00:00 +0: Unsupported platform shows a recoverable setup screen
00:00 +1: All tests passed!
```

Figure 1 Static analysis and setup widget test passed

Campus Connect | Connected targets

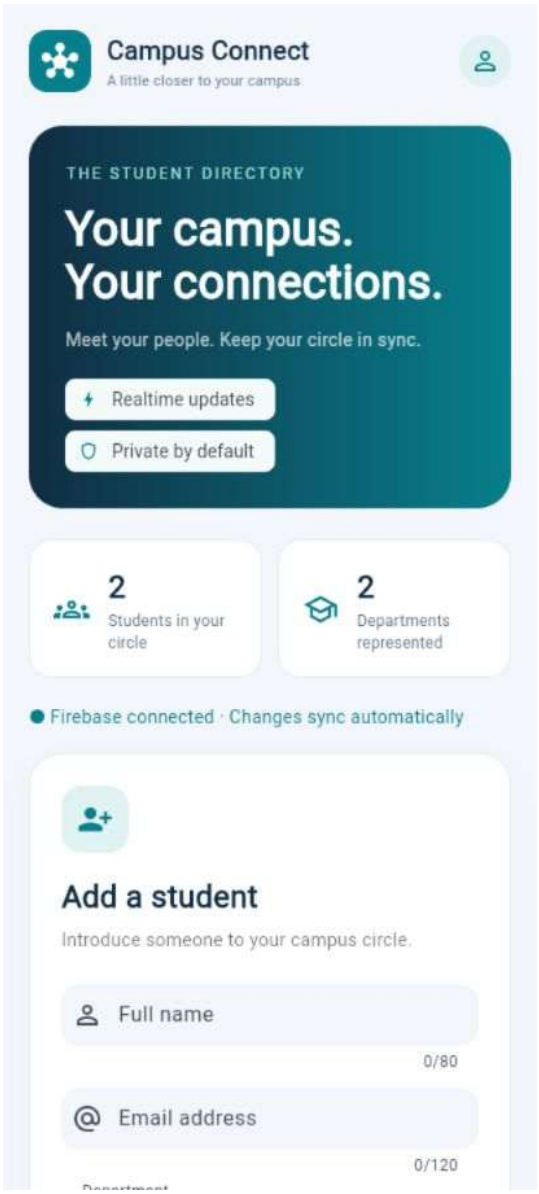
```
PS campus_connect> flutter devices
Found 3 connected devices:
  Windows (desktop) - windows - windows-x64    - Microsoft Windows [Version 10.0.26200.9457]
  Chrome (web)      - chrome  - web-javascript - Google Chrome 154.0.8037.98
  Edge (web)        - edge    - web-javascript - Microsoft Edge 155.0.4283.45
```

Figure 2 Device inventory from flutter devices

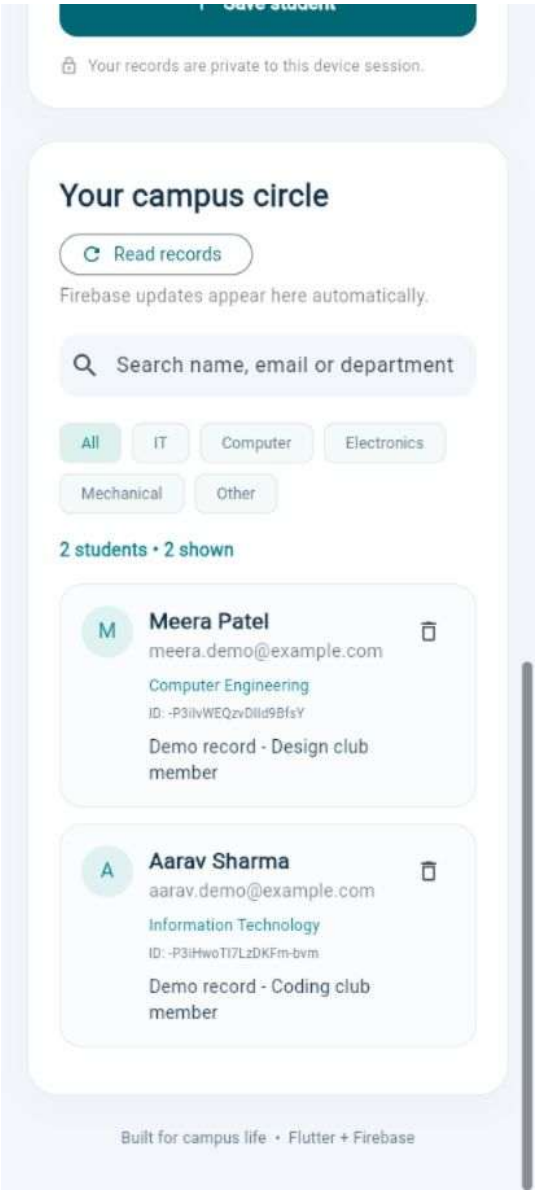
Static analysis reported no issues and the setup widget test passed. The available targets were Windows, Chrome, and Edge. No Android emulator or physical device was connected.

Output Mobile User Interface

Actual Campus Connect screenshots captured at a 412 by 915 pixel mobile viewport. The app is running as a mobile web preview with a live Firebase connection.



Dashboard and connection status



Student directory and read action

Figure 3 Mobile layout showing two saved students and realtime Firebase status

The mobile interface stacks the input form and directory vertically. Student records were saved and retrieved from Firebase and remained visible after reloading the app.

Output Keystore Creation and Certificate

A real upload keystore was created at C:\Users\Parth\.android\campus-connect-upload.jks. Certificate details were verified using keytool. The terminal images do not display the password or private key.

```
keytool -genkeypair -v -keystore <keystore-path>
  -storetype JKS -keyalg RSA -keysize 2048
  -validity 10000 -alias upload
keytool -list -v -keystore <keystore-path> -alias upload
```

Campus Connect | Keytool certificate output

```
PS> keytool -genkeypair ... -alias upload
Generating 2,048 bit RSA key pair and self-signed certificate (SHA384withRSA) with a validity of
10,000 days
    for: CN=Parth Chaudhari, OU=Campus Connect
[Storing C:\Users\Parth\.android\campus-connect-upload.jks]


PS> keytool -list -v ... -alias upload
Alias name: upload
Creation date: 12-Oct-2026
Entry type: PrivateKeyEntry
Certificate chain length: 1
Certificate[1]:
Owner: CN=Parth Chaudhari, OU=Campus Connect
Issuer: CN=Parth Chaudhari, OU=Campus Connect
Serial number: cce14531eb46b36a
Valid from: Mon Oct 12 04:36:40 IST 2026 until: Fri Feb 27 04:36:40 IST 2054
Certificate fingerprints:
     SHA1: 77:EE:EB:B1:84:87:6E:7A:80:BB:48:E9:2C:9C:BB:7C:0D:8E:5F:56
     SHA256:
CF:F2:F6:21:3E:18:69:BC:1C:0D:9D:54:27:88:01:5F:54:E8:05:31:48:4E:26:CD:90:DA:DF:39:9C:C6:98:C2
Signature algorithm name: SHA384withRSA
Subject Public Key Algorithm: 2048-bit RSA key
Version: 3
```

Figure 4 Actual key generation and certificate verification output

Output Signing Configuration

The keystore alias is upload. The local properties file is excluded by android/.gitignore. The report copy below hides both passwords.

android/key.properties | Passwords hidden

```
storePassword=[HIDDEN]
keyPassword=[HIDDEN]
keyAlias=upload
storeFile=C:\\Users\\Parth\\.android\\campus-connect-upload.jks
```

Figure 5 key.properties with passwords hidden

The Gradle release build now uses the release signing configuration instead of the development debug key.

android/app/build.gradle.kts | Release signing

```
signingConfigs {
        create("release") {
            keyAlias = keystoreProperties.getProperty("keyAlias")
            keyPassword = keystoreProperties.getProperty("keyPassword")
            storeFile = keystoreProperties.getProperty("storeFile")?.let { file(it) }
            storePassword = keystoreProperties.getProperty("storePassword")
        }
    }

    buildTypes {
        release {
            signingConfig = signingConfigs.getByName("release")
        }
    }
```

Figure 6 Applied signing configuration in build.gradle.kts

Campus Connect | Android toolchain check

```
PS campus_connect> flutter doctor -v
Android toolchain - develop for Android devices (Android SDK version 36.0.0) [374ms]
    - Android SDK at C:\Users\Parth\AppData\Local\Android\sdk
    - Emulator version 37.1.11.0 (build_id 15917651) (CL:N/A)
    X cmdline-tools component is missing.
      Try installing or updating Android Studio.
      Alternatively, download the tools from https://developer.android.com/studio#command-line-
tools-only and make sure to set the ANDROID_HOME environment variable.
      See https://developer.android.com/studio/command-line for more details.
    X Android license status unknown.
      Run `flutter doctor --android-licenses` to accept the SDK licenses.
      See https://flutter.dev/to/windows-android-setup for more details.
```

Figure 7 Android toolchain diagnostic output

Campus Connect | Release APK build attempt

```
PS campus_connect> flutter build apk --release
Running Gradle task 'assembleRelease'...
WARNING: A restricted method in java.lang.System has been called
```

Figure 8 Actual Gradle build progress before the attempt was stopped

The Android SDK is detected, but command-line tools are missing and license status is unknown. The APK build reached assembleRelease and was stopped before completion. No completed APK has been verified. Android device testing and App Bundle verification remain pending.

```
flutter build apk --release
flutter build appbundle --release

Expected outputs after successful builds
build/app/outputs/flutter-apk/app-release.apk
build/app/outputs/bundle/release/app-release.aab
```

Conclusion

Campus Connect passed static analysis and its setup widget test. Its mobile web interface demonstrated a live Firebase connection and persistent student records. An RSA upload keystore was created and verified, and the Gradle release signing configuration was applied. The Android toolchain must be completed and a signed build validated on an Android device before deployment.